

Disentangling the role of subsurface storage in the propagation of drought through the hydrological cycle

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ABSTRACT

Subsurface storage changes (ΔS) represent a key modulator of drought propagation through the hydrological cycle, but their contribution to the annual water balance, and to drought propagation and recovery has rarely been explicitly assessed across catchments and climates. To expand on previous work on this matter, here we performed a large-sample analysis of precipitation, discharge, actual evapotranspiration (ET), and ΔS for 10 hydrological years and 102 catchments across various hydro-climatological regimes in Italy. We found that ΔS cannot be neglected in the annual water balance. Storage depletion leads to the attenuation of hydrological drought compared to meteorological drought, meaning that subsurface storage actively supports discharge during drought. We also found that storage generally recovers from precipitation deficits over time scales similar to the discharge recovery time, while recovery times for ET are longer. These findings show that subsurface storage drives drought propagation and recovery, regardless of climatic and catchment characteristics, and are thus relevant to properly inform water managers about surface- and ground-water availability in a changing climate.

1. Introduction

Drought is among the most impactful natural hazards in term of affected people all over the world (CRED, 2020) and has severe and intertwined impacts on environmental, economical, and social aspects, such as terrestrial and freshwater ecosystems, agriculture, water supply, human health, and social stability (Stephan et al., 2021). Furthermore, drought hazard is expected to increase in frequency, magnitude, and duration in the near future due to climate change (Samaniego et al., 2018; Cammalleri et al., 2020; Haile et al., 2020).

Drought starts from a deficit in precipitation - P -, possibly in combination with an increase in temperature (meteorological drought); a meteorological drought then propagates through the hydrological cycle to generate soil moisture deficits (soil moisture or agricultural drought), as well as discharge - Q - and groundwater deficits (hydrological drought) (Van Loon, 2015). Here we use the term hydrological drought to refer to discharge deficits. The process leading from meteorological to hydrological drought, which is known as drought propagation, is governed by local climatic and topographic features (Van Loon and Laaha, 2015), along with human influences (Van Loon et al.,

2016, 2022). Hydrological drought is generally attenuated compared to the corresponding meteorological drought (Van Loon and Van Lanen, 2012) by land surface processes, i.e., actual evapotranspiration - ET - and catchment storage. However, prolonged dry periods can also lead to an intensification of hydrological drought (Saft et al., 2015; Garreaud et al., 2017; Tian et al., 2018; Avanzi et al., 2020; Alvarez-Garreton et al., 2021; Maurer et al., 2022; Massari et al., 2022). Also the period required to recover from deficit conditions (drought recovery) may vary across the different components of the hydrological cycle (Parry et al., 2016a), and it is also affected by catchment properties (Parry et al., 2016b) and human factors (Margariti et al., 2019; Wu et al., 2019). Thus, quantifying water availability in each hydrological compartment during and after dry periods still remains challenging because of several factors involved in drought propagation and recovery, but it is essential to properly inform water managers.

A growing number of papers studied drought propagation (e.g., Van Loon and Van Lanen (2012), Lorenzo-Lacruz et al. (2013) and Wu et al. (2021a)) and recovery (e.g., Wu et al. (2021b)), as well as

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the influence of climatic (Apurv et al., 2017) and catchment properties (Van Loon and Laaha, 2015; Barker et al., 2016; Apurv and Cai, 2020; Parry et al., 2016b; Yang et al., 2017) on them. Previous works mostly used the threshold level method (Van Loon and Laaha, 2015; Apurv and Cai, 2020; Parry et al., 2016b; Yang et al., 2017) or standardized indices (Barker et al., 2016; Apurv et al., 2017) to identify meteorological and hydrological droughts, and to characterize them in term of drought characteristics (severity and duration). More importantly, they used catchment properties and hydrological signatures to represent the land surface processes involved in drought propagation. For instance, Van Loon and Laaha (2015) studied how various physiographic catchment properties control hydrological drought characteristics for 44 Austrian catchments and they found that catchment storage, represented by the Base Flow Index (BFI), affects hydrological drought duration. Further, Barker et al. (2016) analysed drought propagation for 121 catchments in UK, showing that catchment properties related to storage, such as the BFI, the percentage of productive aquifer in the catchment, and the soil wetness, control hydrological drought characteristics, as well as its delay with respect to the meteorological drought.

However, few studies focused on explicitly quantifying each hydrological flux and storage to provide mechanistic understanding of hydrological processes during drought propagation and recovery (Teuling et al., 2013; Bales et al., 2018; Soulsby et al., 2021). Teuling et al. (2013) showed that ET aggravated soil moisture deficits during the 2003 European summer drought across a set of experimental catchments, while Soulsby et al. (2021) found that subsurface storage sustained Q in an experimental Scottish catchment during the Northern European 2018 drought and the following recovery. Both the processes revealed by Teuling et al. (2013) and Soulsby et al. (2021) lead to a net depletion of subsurface storage during drought (Safeeq et al., 2021), which in turn can lead to a reduction of available groundwater for human purposes. Thus, we argue here that considering the whole hydrological cycle is necessary to better understand the hydrological processes occurring during drought propagation and recovery, and to quantify water availability in each hydrological compartment during such periods.

To pursue such an approach, we rely on the water balance model (Eq. (1)) following Bales et al. (2018).

$$P = Q + ET + \Delta S \quad (1)$$

Here we define ΔS as the subsurface storage change (hereafter subsurface storage) in the whole regolith, either in the soil and in the weathered bedrock. Storage changes could also be due to surface water bodies, snowpack, and glaciers within the catchment. Nonetheless, the annual time scale and the selection of suitable catchments (non-glacierized and not regulated by lakes, for instance) allowed us to minimize snow influence and the effects due to glaciers and surface water bodies. Therefore, at annual time scale positive annual ΔS values represent an inter-annual carryover in the subsurface storage, while negative values an inter-annual drawdown. The assumption that catchments do not show any inter-annual change in subsurface storage may not hold true (Bouaziz et al., 2018; Fan, 2019; Wlostowski et al., 2021; Han et al., 2020), due to net intercachment groundwater flow – that can be part of ΔS – or catchment properties such as large depth of soil and heavily weathered bedrock, for which we expect large subsurface storage changes and variability.

Catchment-scale water balance datasets have historically been challenging to collect, especially for comparative studies across a variety of catchments. Ground-based ET and ΔS (soil moisture and groundwater) data are seldom available and mostly limited to small-sized experimental catchments, while total terrestrial water storage estimates from GRACE satellites can be used only for large-sized catchments, due to their coarse spatial resolution (Landerer and Swenson, 2012). Consequently, systematic large-sample analyses across a wide range of catchments to quantify ΔS contribution to the allocation of P through

the hydrological cycle (Wang et al., 2014; Safeeq et al., 2021) and to drought propagation and recovery are rare, despite few exceptions (Orth and Destouni, 2018). In order to fill this gap, here we hypothesized that blending ground-based and remote-sensed data can provide a consistent, large-scale water balance dataset across different climates and landscapes.

Thus, in this work we aimed at answering three research questions: (i) how much is the interannual carryover or drawdown in subsurface storage for a large-set of Italian catchments across different hydroclimatic regimes?, (ii) does this subsurface storage carryover or drawdown affect drought propagation?, and (iii) do ΔS , ET, and Q recover over similar time scales after a deficit in P? To answer these questions, we leveraged the dataset we collected to perform a large-sample analysis of each water balance component for 102 Italian catchments over the period 2010–2019 and gain insights into the propagation of meteorological drought through the hydrological cycle for annual drought events in the considered decade.

2. Material and methods

We chose Italy as study region because it presents a broad variety of climatic and topographic features, as briefly described in Bruno et al. (2021). Moreover, it experienced several drought episodes over the recent years (Stahl et al., 2016; Chiogna et al., 2018; Spinoni et al., 2019), such as the 2012 and 2017 events, and thus it is a suitable study region for the objectives of the work.

Here, we selected the period from hydrological year (h.y.) 2010 to h.y. 2019 as study period, and we considered h.y. as the period from September to August (Bruno et al., 2021).

2.1. Study catchments

We collected estimates for each water balance component (Eq. (1)), as well as hydro-meteorological data and catchment properties for 102 partly-nested catchments across Italy (Fig. 1). We carried out catchment delineation using the Shuttle Radar Topographic Mission (SRTM) Digital Elevation Model (DEM) and we used catchment boundaries from it to derive catchment-scale hydro-meteorological data and properties (see Section 2.2 for details about each of them). We selected the study catchments according to some criteria to make them suitable for the analysis: (i) catchment area greater than 100 km², as the spatial resolution of the selected satellite-derived ET product is around 4 km × 5 km (Section 2.2); (ii) annual runoff ratio less than 1.5, as an arbitrarily chosen threshold to guarantee coherence between the P and Q data, but at the same time to allow for possible import from the subsurface storage; (iii) percentage increase in catchment area greater than the 10% between upstream and downstream nested catchments, to avoid clusters of nested catchments (Alfieri et al., 2020) that could bias our general results; (iv) relative difference between the catchment area we obtained from SRTM DEM and the one provided by regional hydrometeorological offices – if available – less than 20%, to exclude errors in our catchment delineation (Alfieri et al., 2020); (v) glacier cover, from the RGIv6.0 glacier outlines (RGI, 2017), less than 10% of catchment area to avoid glacierized catchments (Alvarez-Garretton et al., 2021) and effects in the drought propagation owing to glacier contribution to storage (Van Tiel et al., 2021).

The resulting study catchments (Fig. 1 and Table B.1) cover different hydro-climatic regimes – as identified by the aridity index (AI) introduced in Section 2.2 –, a steep orographic gradient, and include medium-to-large size catchments. Energy-limited catchments (AI < 1) are the 24% of study catchments, while very energy-limited catchments (AI < 0.75) are the 53% of them. Further, water-limited catchments (AI > 1) are the 17% of study catchments, while very water-limited catchments (AI > 1.25) are the remaining 6%, these latter mostly in the central and southern regions of Italy. Catchment elevation has a mean (min/max) value of 1076 (143/2487) m a.s.l., while the mean (min/max) catchment area is 3631 (105/68619) km². Moreover, the study catchments are characterized by different dominant soil type, land cover, and geological classes (Table B.1).

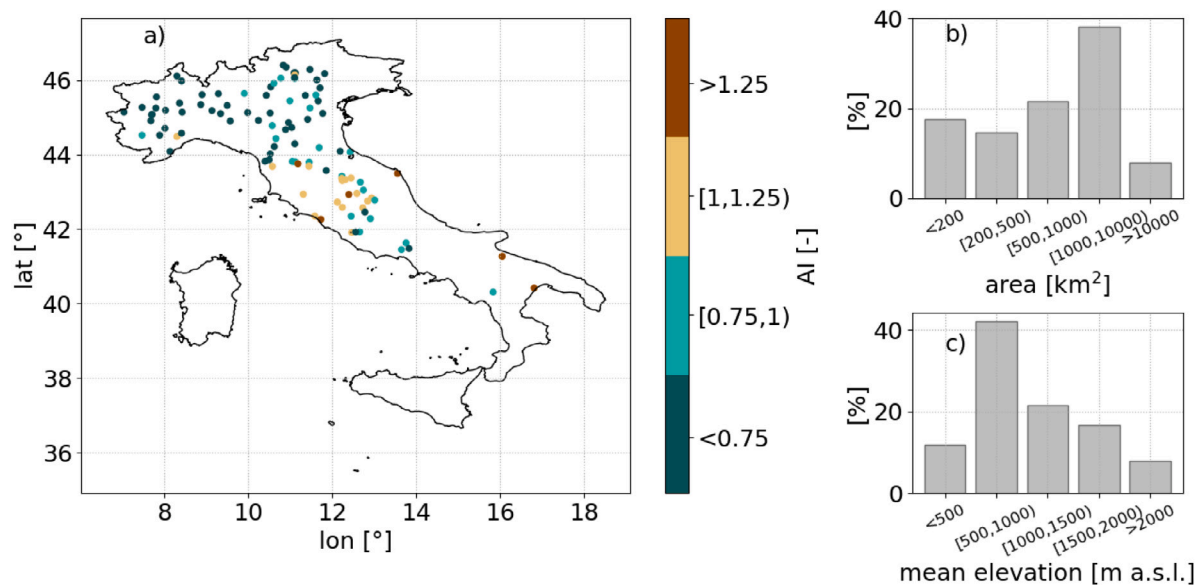


Fig. 1. Location of the closure section and properties for the study catchments: (a) aridity index, AI for each catchment; (b) frequency distribution of catchment area; (c) frequency distribution of catchment mean elevation.

Table 1

Overview of the datasets used in the study.

Variable	Dataset	Reference	Source	Purpose
T	DPC	Ghilain et al. (2011), EUMETSAT (2016)	https://landsaf.ipma.pt/en/products/evapotranspiration-energy-flxs/met/ (last access, 24 September 2020)	PET estimation
P	DPC			Water balance analysis
Q	DPC			Water balance analysis
ET	LSASAF (METv2)			Water balance analysis
P	BIGBANGv4.0	Braca and Ducci (2018), Braca et al. (2019), ISPRA (2021)	https://groupware.sinanet.isprambiente.it/bigbang-data/library/bigbang40/grids (last access, 27 April 2022)	Multi-dataset comparison and SPI calculation
ET	BIGBANGv4.0	Braca and Ducci (2018), Braca et al. (2019), ISPRA (2021)	https://groupware.sinanet.isprambiente.it/bigbang-data/library/bigbang40/grids (last access, 27 April 2022)	Multi-dataset comparison
ET	FLUXNET2015	Pastorello et al. (2020)	https://fluxnet.org/data/fluxnet2015-dataset/ (last access, 20 April 2020)	Multi-dataset comparison
ET	GLEAMv3.3a	Miralles et al. (2011)	https://land.copernicus.eu/pan-european/corine-land-cover/clc2018 (last access, 22 November 2020)	Multi-dataset comparison
Digital Elevation Model	STRM DEM	RGI (2017)		Catchment characterization
Glacier outlines	RGIv6.0			Catchment characterization
Land cover	Corine Land Cover 2018			Catchment characterization
Soil properties	European Soil Database Derived data	Hiederer (2013a,b)	http://www.sinanet.isprambiente.it/it/sia-ispra/download-mais/complessi-idrogeologici/view (last access, 02 August 2021)	Catchment characterization
Geological properties	ISPRA hydro-geological complexes			Catchment characterization

2.2. Data

2.2.1. Water balance dataset

To perform a large-sample analysis across different hydroclimatic regimes, we collected data from multiple sources to close the annual water balance by blending ground-based measurements and remote sensing. We further collected alternative estimates of P and ET, as detailed in the following paragraphs and Table 1, for a multi-dataset comparison to verify the consistency of the water balance dataset.

We derived daily catchment-average precipitation from ground-based P data from the Italian regional hydrometeorological offices

and the Italian Civil Protection Department (in the following, DPC dataset) (Bruno et al., 2021). Additionally, we used monthly catchment-average P data from the BIGBANGv4.0 dataset (Braca and Ducci, 2018; Braca et al., 2019; ISPRA, 2021). BIGBANG is a national-scale gridded water balance model, relying on ground-based P and temperature (T) data and parsimonious formulations. It provides estimates of each water balance component (precipitation, actual evapotranspiration, surface flow, change in soil storage, groundwater recharge) along with potential evapotranspiration (PET), at monthly and 1 km resolutions for the Italian territory over the period 1951–2019 (Braca and Ducci, 2018;

Braca et al., 2019; ISPRA, 2021). We used BIGBANG P data for comparison with P estimates from the DPC dataset and for the calculation of the Standardized Precipitation Index (Section 2.3) (McKee et al., 1993), due to its long temporal availability.

For discharge, we relied on data from the DPC dataset (Bruno et al., 2021) and the Italian regional hydrometeorological offices. We further applied some quality checks to the Q dataset to identify and remove possible outliers: (i) a Hampel filter (Hampel, 1974), which compares each value with statistics from a surrounding window, and (ii) a filter similar to the one used in Gudmundsson et al. (2018), comparing each value with statistics for that day of the year over the entire series. The first filter allowed us to detect individual suspicious values, due to temporary malfunctioning in the measurement instrumentation for instance, while the second one enabled us to identify consecutive values outside the climatology for that period of the year, e.g. due to prolonged instrumentation malfunctioning. Thus, we obtained a quality-checked daily mean Q dataset for 102 catchments across Italy (see Fig. D.1 for an example of filtered data). We further underline that Q data for some catchments may be influenced by human activities (e.g., reservoirs and irrigation), but the aim of this study is a large-sample analysis for which a certain degree of anthropogenic influence can be accepted (Murphy et al., 2013) (see Section 4.2 for further discussion).

We derived ET maps for the study region from the already validated METv2 product by the Land Surface Analysis of the EUMETSAT Satellite Application Facility (hereafter, LSASAF product) (Ghilain et al., 2011; EUMETSAT, 2016). The LSASAF product provides sub-daily ET estimates at a spatial resolution of 3.1 km $\times$ 3.1 km at the nadir (around 4 km $\times$ 5 km over Europe), by resolving the surface energy balance from satellite-derived data of radiative forcings, land cover, biophysical parameters, and soil moisture status, along with ancillary meteorological data. Previous works validated it against ground-based flux towers data across Europe (Hu et al., 2015; Petropoulos et al., 2016) and also compared it to the MODIS ET product (Hu et al., 2015). Furthermore, it has been used for agricultural drought monitoring (Sepulcre-Canto et al., 2014) and as benchmark for ET simulated by hydrological modelling in European regions (Bouaziz et al., 2018). To further verify the reliability of the LSASAF product over the study area and period, we compared ET data at point level against eddy covariance flux tower ET data from the FLUXNET2015 dataset (Pastorello et al., 2020). The FLUXNET2015 dataset collects data from flux tower sites across the globe at a half-hourly time resolution, up to December 2014. For the comparison, we selected the sites with more than 80% of data over the comparison period. We also compared the LSASAF product at catchment-scale against two additional gridded ET products, GLEAMv3.3a and the BIGBANG dataset. GLEAM provides global ET and PET estimates at daily temporal resolution and 0.25° spatial resolution from a physically-based land-surface model and satellite-based data (Miralles et al., 2011; Martens et al., 2017).

We assessed ΔS as annual residual from P, Q, and ET data accumulated over the h.y., according to Eq. (1). For comparison, we also derived catchment-scale ΔS estimates using P and ET from the BIGBANG dataset, and the Q data described above. In Section 4.2, we provide a discussion of uncertainties in the data used to close the annual water balance and therefore in ΔS assessment. Moreover, we performed a correlation analysis (Pearson correlation coefficients, r and significance level equal to 0.05) between annual ΔS statistics (mean and standard deviation over the study period) and catchment properties (Section 2.2.2) to test if ΔS variability among the study catchments can be explained by such factors.

2.2.2. Catchment characteristics

To study drought propagation across multiple climatic and morphologic features, we compiled a set of catchment properties to characterize the study catchments in terms of climatic, topographic, land use, pedological, and geological properties.

For climate characterization, we firstly evaluated catchment-average PET through the temperature-based Hamon method (Dingman, 2002) and ground-based T data from the DPC dataset (Bruno et al., 2021). The Hamon method is based on temperature data only, but we preferred it to alternative formulations considering also radiation and other meteorological variables in order not to propagate uncertainties related to the spatial interpolation of sparse data as meteorological data can be at a large scale. Further, in this study PET was only used for climate characterization of the study catchments and the drought events, while actual evapotranspiration estimates from the LSASAF product were exploited to study drought evolution across the water balance components. We indeed used PET data to derive the aridity index (AI, Eq. (2) (Maurer et al., 2022)) and so, to characterize the hydro-climatic regime of each catchment (water-limited if $AI < 1$ or energy-limited if $AI > 1$).

$$AI = \frac{PET}{P - \Delta S} \quad (2)$$

Moreover, we estimated the Asynchronicity Index (ASI) between P and PET (Feng et al., 2019) (Eq. (A.1) to (A.7)) as a seasonality index and as an indicator of dry catchments. ASI is an information theory-based and non-parametric index that quantifies the difference in relative magnitude and phase between the P and PET signals (Feng et al., 2019). ASI varies between 0 and 1, and values greater than 0.36 represent Mediterranean regions with de-synchronized annual cycles of P and PET, both in term of relative magnitude and phase shift (Feng et al., 2019). We chose this metric instead of other seasonality indexes as it does not require the sinusoidality assumption for P, which could not hold true for study catchments in the alpine region (Crespi et al., 2018).

As topographic properties, we computed catchment area, and mean catchment latitude and elevation from the SRTM DEM. We also estimated the dominant land cover class and the percentage of urban, crop, shrub, forest, grass, bare soil, and water bodies from the Corine Land Cover 2018. Additionally, we derived the dominant subsoil and topsoil texture, and the catchment-average percentage of silt, sand, clay, organic carbon, and gravel from the European Soil Database Derived data product (Hiederer, 2013a,b). We then retrieved the dominant geological class and the percentage of carbonate aquifers, as a proxy for karstic systems in the catchment (Fan, 2019), from a national-scale map of hydro-geological complexes.

Finally, we quantified the BFI and the recession coefficient using Koffler and Laaha (2013) which are often used as indicators for catchment memory and groundwater contribution to the discharge (Van Loon and Laaha, 2015; Sutanto and Van Lanen, 2022), as well as the discharge sensitivity to storage as the ratio between the standard deviations of baseflow and hydraulic storage (Apurv and Cai, 2020). To derive baseflow, we used the standard method proposed by Ladson et al. (2013) for baseflow separation, basing on a three-passes digital recursive filter applied to the daily Q data. Furthermore, we computed hydraulic storage via recession analysis following Kirchner (2009) and Wlostowski et al. (2021), from daily Q and ET data.

2.3. Drought propagation analysis

We based our analysis of drought propagation on two steps: (i) drought identification and (ii) drought characterization, as depicted in Fig. 2.

Firstly, for each catchment we identified the drought years for the following analysis (drought identification, Fig. 2a). For this purpose, we computed the Standardized Precipitation Index (SPI) (McKee et al., 1993), based on monthly catchment-average P data over the period 1951–2019 from the BIGBANG P product. SPI is a widely used drought index which only needs long-term (>30 years) monthly P data and provides the probability of a given P anomaly, as deviation from the long-term mean (McKee et al., 1993). In the SPI calculation, historical monthly P data are accumulated over different time scales of interest for drought monitoring (usually from 1 to 48 months) and fitted to a

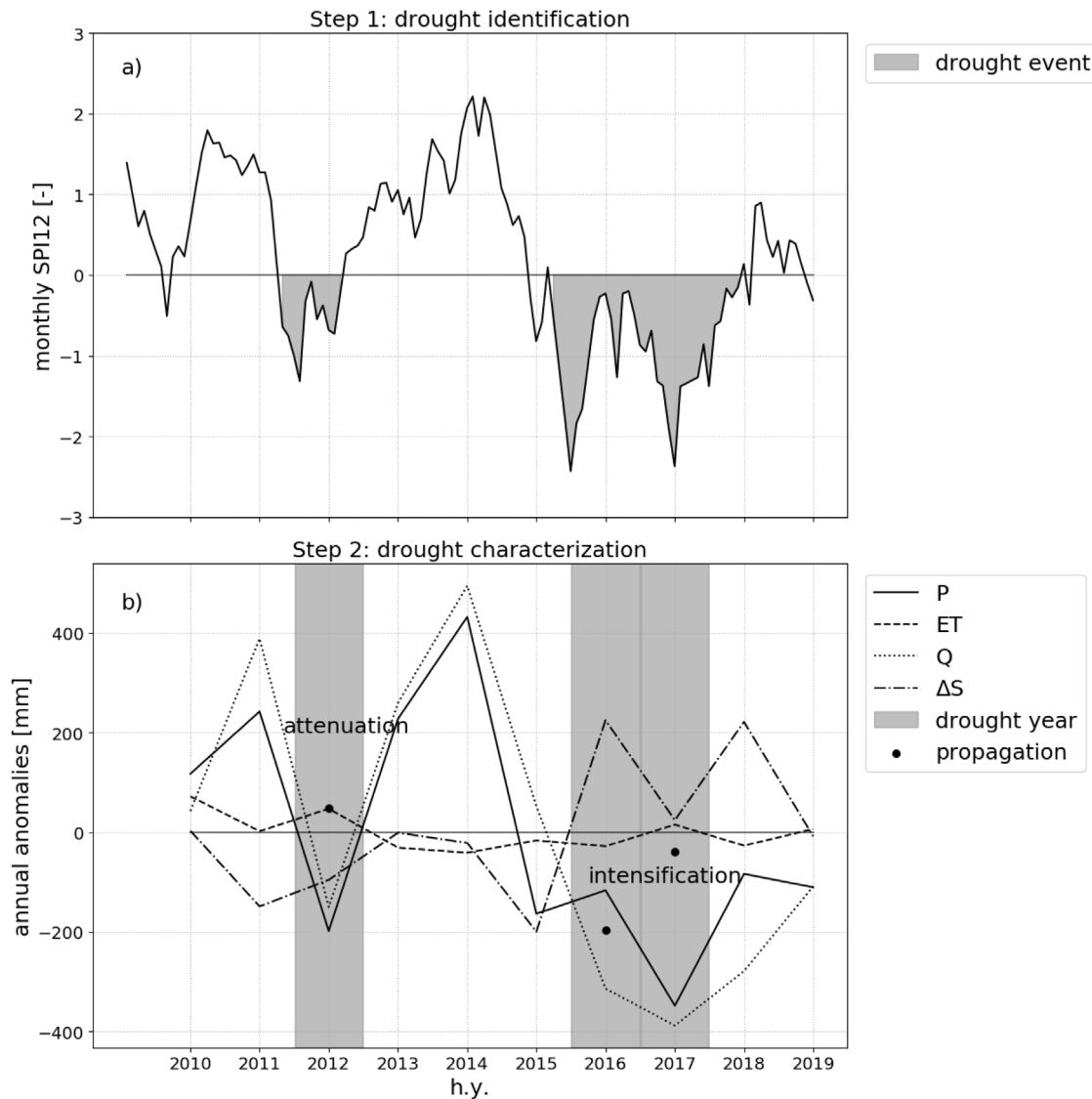


Fig. 2. Illustration of working steps for drought propagation analysis: (1) calculation of the monthly Standardized Precipitation Index with an accumulation period of 12 months (SPI12) for drought identification (a); (2) computation of annual anomalies in precipitation (P), actual evapotranspiration (ET), discharge (Q), and change in subsurface storage (ΔS) and quantification of drought propagation as difference between Q and P anomalies for drought characterization (b).

parametric statistical distribution; the probabilities obtained from the distribution are then transformed into a normal distribution with mean equal to 0 and standard deviation equal to 1 (McKee et al., 1993). A drought event is defined as a period with SPI values less than -1 and identified by the first and the last time steps with consecutive negative values (McKee et al., 1993). Here we chose an accumulation period of 12 months for the SPI calculation (hereafter SPI12), as we focused on the propagation from meteorological to hydrological drought and SPI12 allows the detection of long periods of P deficit that could lead to hydrological drought (Apurv et al., 2017; Garreaud et al., 2017; Tian et al., 2018). Following Stagge et al. (2015), we hypothesized a number of candidate distributions for the SPI12 calculation and verified the normality for the resulting values through a Shapiro–Wilk test (p -value < 0.05). Thus, we used the Gamma distribution as the one from which we obtained the lower Shapiro–Wilk rejection frequency across the study catchments (Table C.1). We considered hydrological years as drought years if two criteria were met: (i) SPI12 of the last month of the h.y. following a drought event as defined above and (ii) negative P annual anomaly, as defined later, according to the P data from the DPC dataset. The latter criteria allowed us to verify the consistency

between the two P datasets used in the study (i.e., the DPC dataset for the analyses and the BIGBANG dataset for drought identification).

Secondly, we characterized drought years in terms of drought severity and attenuation or intensification of the hydrological drought (drought characterization, Fig. 2b). Thus, we computed P, Q, ET, and ΔS annual anomalies, as departure from mean annual values. Throughout the study, we considered P anomaly as a metric for the severity of the meteorological drought, while Q anomaly for the severity of the hydrological one. Further, we defined a drought propagation measure, as the difference between Q and P annual anomalies, similarly to Alvarez-Garretón et al. (2021). A positive drought propagation value stands for an attenuation of the hydrological drought compared to the meteorological, while a negative drought propagation value for an intensification.

3. Results

3.1. Annual water balance components

Catchment-average mean annual P ranges from 717 mm to 1764 mm and annual P standard deviation from 60 mm to 353 mm. Their variability across the study area reflects climate and orography (Fig. 3a).

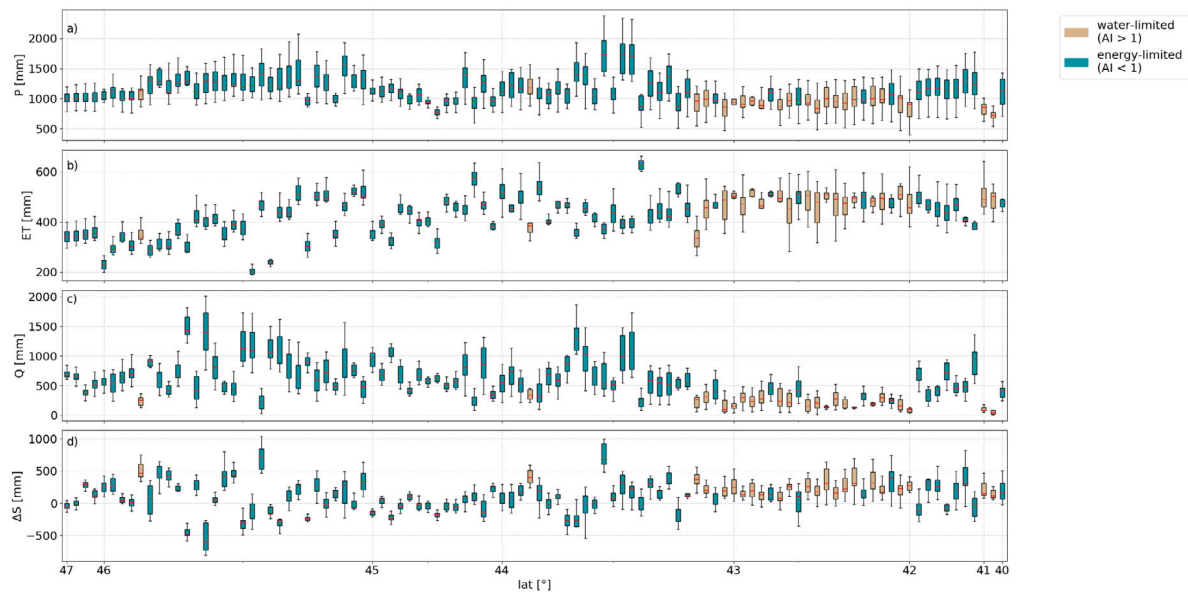


Fig. 3. Boxplot of each annual water balance component over the study period for each catchment (north to south ordered, from the left to the right hand side): (a) precipitation, P; (b) actual evapotranspiration, ET; (c) change in subsurface storage, ΔS ; (d) discharge, Q. Each boxplot is colour coded according to the catchment aridity index, AI.

For instance, some energy-limited catchments at around 44° latitude – corresponding to catchments along the northern-western coast of Italy in the upper Tuscany region – are characterized by higher annual P amounts than the surrounding energy-limited catchments, due to the complex orography in that area (Crespi et al., 2018).

Catchment-average mean annual ET spans from 206 mm to 637 mm (Fig. 3b) and annual ET standard deviation from 16 mm to 90 mm. ET shows higher mean values and inter-annual variability in the water-limited central and southern catchments than in the energy-limited northern catchments. On average mean annual ET consists in 41% of P across the study catchments (17%–77% of P, as minimum and maximum).

Mean annual Q varies from 74 mm to 1503 mm and annual Q standard deviation from 23 mm to 428 mm (Fig. 3c). Q has a contrasting behaviour compared to ET, with higher mean annual values and inter-annual variability in the northern energy-limited catchments. Mean annual Q average contribution to P across the study catchments is 48%.

The mean annual ΔS ranges from -534 mm to 747 mm (Fig. 3d) and annual ΔS standard deviation from 52 mm to 274 mm. Mean annual ΔS thus is on average (min/max) 11% (-41% – 50%) of the corresponding P across all catchments. Furthermore, the maximum standard deviation of annual ΔS corresponds to 24% of P, while maximum standard deviations of annual ET and Q are 20% of P and 23% of P. Thus, ΔS shows a slightly higher variability than the other water balance components, when compared to P.

The correlation analysis between ΔS statistics over the study period and possible predictors revealed that catchment properties can partly explain the differences in ΔS behaviour across the study catchments (Table 2). Catchment properties showing a significant correlation with mean annual ΔS can be thought as predictors for the tendency of the catchment to have an inter-annual subsurface storage carryover or drawdown, whereas the properties showing a significant correlation with the standard deviation of annual ΔS can be predictors for its inter-annual variability. While catchment properties show somewhat low correlations with ΔS statistics ($-0.35 < r < 0.59$), several correlations are consistent with what expected and with previous works (e.g., Bouaziz et al. (2018), Fan (2019), Włostowski et al. (2021) and Han et al. (2020)), as discussed in Section 4.1; this increases confidence in our estimates. For example, catchments with a higher percentage of carbonate aquifers and silty soils, and so a higher storage capacity,

Table 2

Pearson correlation coefficients between change in subsurface storage (ΔS) statistics (mean and standard deviation) and catchment properties (mean catchment latitude, mean catchment elevation, catchment area, percentage of shrub in the catchment, percentage of forest in the catchment, percentage of bare soil in the catchment, percentage of silty soil in the catchment, percentage of carbonate aquifer in the catchment, Asynchronicity Index). Sources and details about each property are provided in Section 2.2.2. Significant coefficients (p -value < 0.05) are reported in bold.

	Lat	Elev	Area	Shrub	Forest	Bare	Silt	Carbonate	ASI
mean annual ΔS	-0.18	-0.2	-0.12	-0.31	0.13	-0.21	0.18	0.23	0.28
std annual ΔS	-0.35	-0.19	-0.3	-0.16	0.33	-0.27	0.37	0.59	0.37

have higher mean annual ΔS ($r = 0.23$ and $r = 0.18$, respectively). Catchments with higher ASI index, and so a marked dry season and a Mediterranean-type climate, have higher mean annual ΔS ($r = 0.28$) and higher standard deviation of annual ΔS ($r = 0.37$). Also, the standard deviation of annual ΔS is negatively correlated with catchment mean latitude ($r = -0.35$) and positively correlated with the percentage of forest, silt, and carbonate aquifers in the catchment ($r = 0.33$, $r = 0.37$, and $r = 0.59$, respectively), along with the recession coefficient we computed from Q data ($r = 0.23$, not shown in Table 2). This suggests that catchments at a higher latitude have a smaller variability in inter-annual ΔS , while catchments with a higher percentage of forests and silty soils, a drier climate, a higher percentage of carbonate aquifers, and a higher groundwater contribution to the discharge have a higher variability of ΔS .

3.2. Water balance response to drought

During the study period, wet and dry years alternated across Italy: h.y. 2012 and 2017 (2013 and 2014) were dry (wet) periods for the whole study area, according to P anomalies (Fig. 4a). Several other dry events occurred across the study catchments, but they were geographically less extensive (e.g., during h.y. 2018, northern catchments experienced negative P anomalies, while the central and southern ones had positive P anomalies). According to SPI12, 178 years across 79 catchments were identified as drought years and major drought events in terms of affected catchments were h.y. 2012 and 2017, in agreement with the literature Stahl et al. (2016), Chiogna et al. (2018) and Spinoni et al. (2019) and drought reports EDO (2022a) and EDO (2022b).

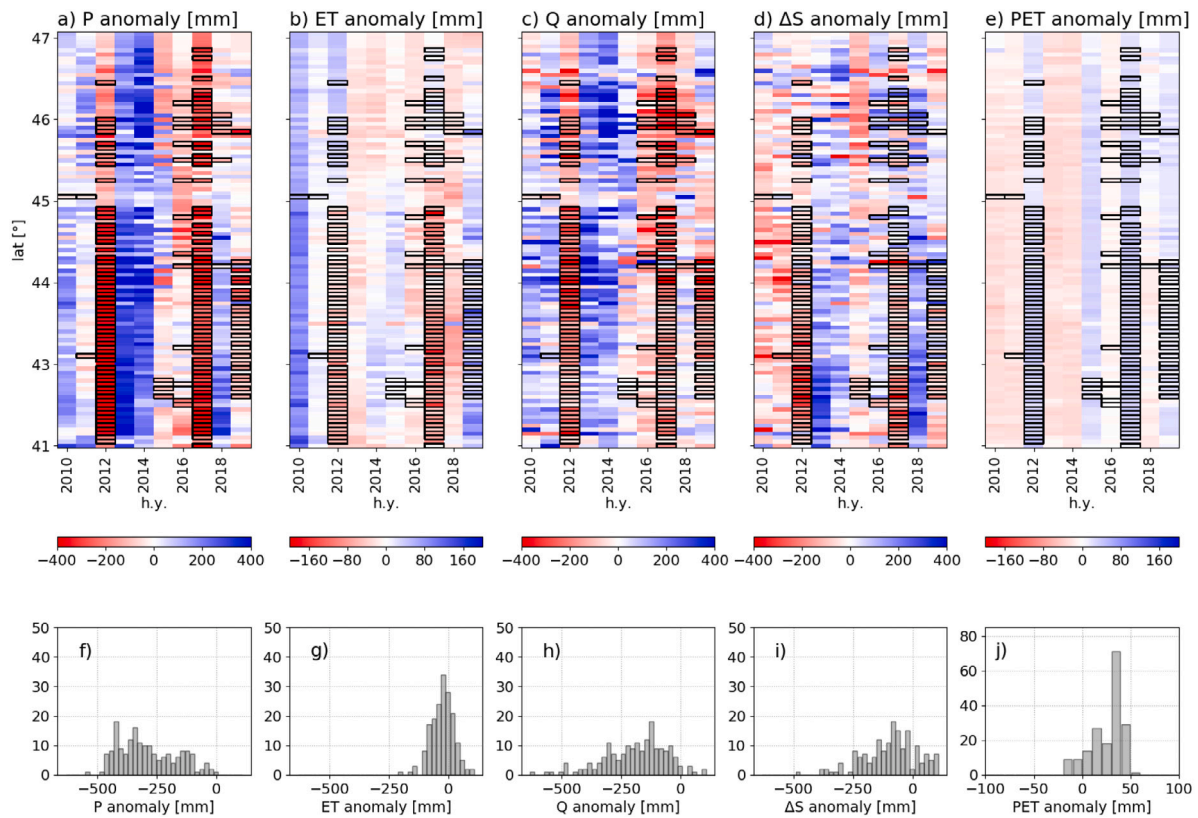


Fig. 4. Annual anomalies for: (a) precipitation, P; (b) actual evapotranspiration, ET; (c) discharge, Q; (d) change in subsurface storage, ΔS ; and (e) potential evapotranspiration, PET over the study period and catchments. Black edges correspond to drought years. Frequency distribution of annual (f) P, (g) ET, (h) Q, (i) ΔS , and (j) PET anomalies during drought years.

Drought years mostly corresponded to dry-warm periods experiencing concurrent P anomalies and positive PET anomalies (median annual P anomaly equal to -303 mm and median annual PET anomaly equal to 31 mm, Fig. 4). During such years, annual anomalies of the four water balance components showed contrasting behaviours across the study catchments (Fig. 4). For both the 2012 and 2017 droughts, ET anomalies were positive for catchments located at latitudes greater than 45° and negative for southern catchments (Fig. 4b); this spatial pattern could reflect the climate and the dominant land cover type in the catchments, as at latitudes greater than 45° catchments are energy-limited, non-Mediterranean, and coniferous-dominated (Table B.1). ΔS and Q anomalies did not show clear distinction between positive and negative values across different hydro-climatological regions in the study area (Fig. 4). During drought years, ΔS anomalies were generally negative (median annual anomaly -77 mm, Fig. 4i) and Q anomalies were generally attenuated compared to P anomalies (median annual P anomaly equal to -303 mm and Q anomaly equal to -178 mm, Fig. 4f and Fig. 4h) across the study catchments.

Focusing on ET, ΔS , and the drought propagation measure, i.e., the difference between Q and P anomalies, negative ΔS anomalies lead to positive propagation values and thus to an attenuation of the hydrological drought (greyish dots in Fig. 5a), while positive ΔS anomalies lead to negative propagation values and so to an intensification of the hydrological drought (reddish dots in Fig. 5a) across the study catchments experiencing drought years over the study period. On the contrary, positive and negative ET anomalies lead to both an attenuation or an intensification of the hydrological drought (Fig. 5a). Thus, ΔS emerges as the key driver of drought propagation through the hydrological cycle.

The drought propagation measure showed statistically different (two-sample Kolmogorov–Smirnov test, p -value < 0.05) distributions

between catchments with contrasting storage-related properties, namely (i) a positive or negative mean annual ΔS (Fig. 6a) and (ii) a positive or negative annual ΔS in the year preceding the drought (Fig. 6b). Distributions are partly overlapping, but catchments with a positive mean annual ΔS had a positive median propagation, and so an attenuation of the hydrological drought, whereas catchments with a negative mean annual ΔS had a slightly negative median propagation, i.e., an intensification of the hydrological drought. Similarly, catchments that in the year preceding the drought had a carryover in subsurface storage ($\Delta S > 0$) had a higher median propagation than catchments that had a drawdown ($\Delta S < 0$). Both these findings indicate that catchments replenishing the subsurface storage, as long-term average characteristic or in the year preceding a drought, can attenuate the hydrological drought.

We also investigated the relationship between the tendency of catchments to attenuate or intensify the drought, quantified via the mean propagation measure during drought years, and climatic and morphological properties of the catchments, such as the ASI and the discharge sensitivity to storage (see Section 2.2.2 for a description of them). A moderate positive linear correlation was found between mean propagation and ASI (Pearson correlation coefficient, $r = 0.43$, p -value < 0.05 , Fig. 7a), while no significant linear correlation was detected between mean propagation and the discharge sensitivity to storage (Fig. 7b). Also the non-linear correlation between mean propagation and the discharge sensitivity to storage was somehow low (Spearman correlation coefficient, $\rho = -0.39$).

3.3. Water balance recovery from drought

The average temporal evolution of P, ET, ΔS , and Q across 55 catchments, which experienced both the 2012 and 2017 droughts,

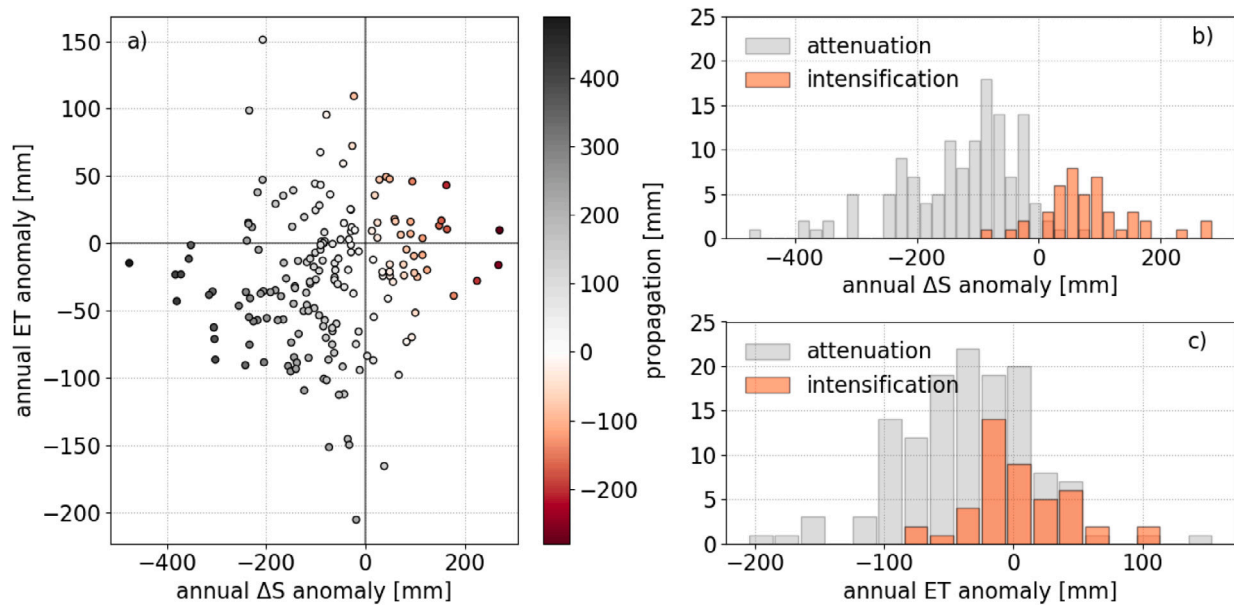


Fig. 5. (a) Scatterplot between annual subsurface storage, ΔS and evapotranspiration, ET anomalies during drought years; dots are coloured according to the propagation measure (difference between discharge and precipitation anomalies). Catchments attenuating the Q anomaly have greyish colour, while catchments intensifying the Q anomaly reddish. Frequency distributions of annual (b) ΔS and (d) ET anomalies for catchments attenuating (grey) and intensifying the Q anomaly (orange) during drought years.

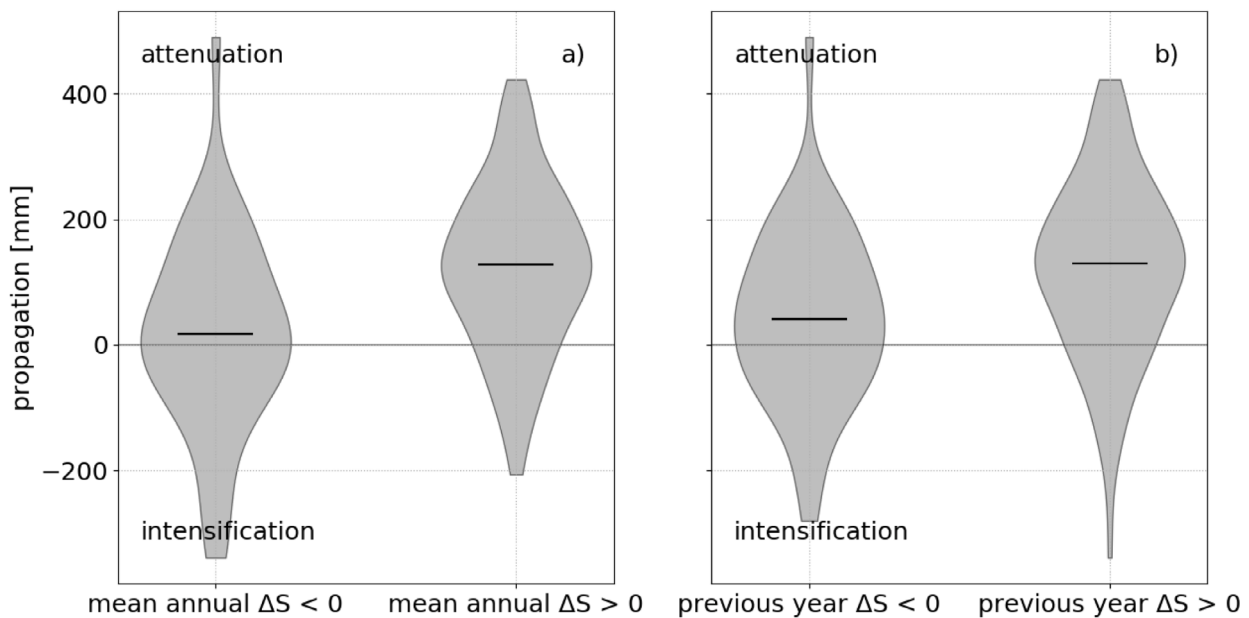


Fig. 6. Distributions of the propagation measure for: (a) catchments with a positive and negative mean annual change in subsurface storage, ΔS and (b) for catchments with a positive and negative ΔS in the year preceding the drought.

and non-drought conditions in the following h.y., revealed different features in the recovery from drought events for the different water balance components. P , Q , and ΔS had a similar timing, both in wet and dry years, while ET was out-of-phase with the P input (Fig. 8). Furthermore, the year following a drought event presented an increase in P , Q , and ΔS values, but still low ET values, even decreasing after the 2012 drought for energy-limited catchments. So, ΔS sustains Q not only during drought, but also during the following recovery period. This sustainment is likely more relevant for gaining rivers (Gleeson and Richter, 2018), which are highly dependent on the groundwater storage.

4. Discussion

In this study we closed the annual water balance for 102 catchments across Italy and 10 hydrological years, and we exploited the dataset to perform a large-sample analysis of the response of each water balance component to drought across different hydro-climatological regimes.

4.1. Main findings

Quantifying catchment-scale water balance is one of the most recurring open questions in hydrology to assess surface and subsurface

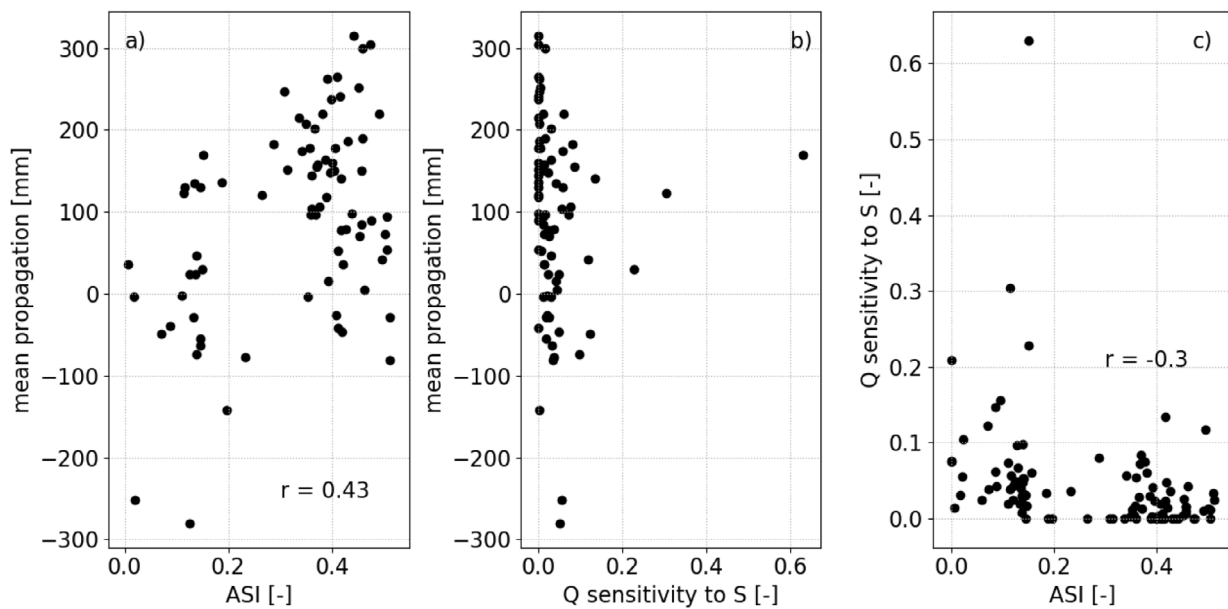


Fig. 7. Scatterplot between: (a) the Asynchronicity Index (ASI) and mean drought propagation; (b) the discharge (Q) sensitivity to storage (S) and mean drought propagation; (c) the ASI and Q sensitivity to S for each catchment experiencing drought years during the study period.

water availability for vegetation, river ecosystems, and human supply (Kampf et al., 2020). Despite advancements in measurement techniques over the last decades, closing the water balance at catchment-scale still remains elusive, mainly because of the paucity of ΔS and ET data, uncertainties in measurements, and spatial heterogeneity within catchments (Kampf et al., 2020). Today, observatories and experimental catchments provide valuable opportunities to assess temporal and spatial scales for which water balance closure is realistic (Safeeq et al., 2021), and remote sensing and hydrological models allow to assess water balance components at large scales, even in data-scarce regions (Abera et al., 2017b).

Assessing water availability across the hydrological compartments during droughts is increasingly important in a warming climate. A water balance perspective can shed light on drought propagation through the hydrological cycle (Teuling et al., 2013; Bales et al., 2018; Orth and Destouni, 2018; Soulsby et al., 2021). Here, we showed that blending ground-based and remote-sensed data can provide a consistent water balance dataset, even at a national scale, to study drought propagation across different climates and landscapes. In this regard, our main findings were three.

First, we provided further empirical evidence that ΔS contribution to the annual catchment-scale water balance cannot be neglected (Han et al., 2020). The average contribution of mean annual ΔS to the long-term water balance is 11% of mean annual P across a large set of temperate catchments (Fig. 3), coherently with previous results for the study region from different approaches (for example Abera et al. (2017a) obtained annual ΔS values ranging from -19% to 5% of P for a prealpine Italian catchment through hydrological modelling). We further showed that ΔS contribution can partly be explained by climate and catchment properties, in agreement with previous studies (e.g., Bouaziz et al. (2018), Fan (2019), Wlostowski et al. (2021) and Han et al. (2020)). For instance, net intercachment groundwater flow, which can be part of ΔS , was demonstrated to be more significant in small catchments with productive aquifers, such as karstified rocks, within the Meuse river basin Bouaziz et al. (2018), while catchments with an arid climate were found to require a longer time to reach a steady state and thus a neglectable contribution of ΔS in the water balance in a large-sample analysis (Han et al., 2020). We indeed found that mean annual ΔS positively correlates with the percentage

of carbonate aquifers in the catchment and the Asynchronicity Index (ASI) between P and PET (Table 2). Moreover, the standard deviation of ΔS negatively correlates with catchment area and positively correlates with ASI (Table 2). These findings illustrate that catchments with a high percentage of carbonate aquifer and a high ASI (i.e., with a dry and Mediterranean-like climate) have a high mean annual carryover in the subsurface storage (or net groundwater export), while catchments characterized by small area and high ASI have a higher variability in annual ΔS than catchments with large area and low ASI values.

Second, we found that the subsurface storage is the key driver in drought propagation across different climates and landscape features, as already revealed for instance by Van Loon and Laaha (2015), Barker et al. (2016) and Apurv and Cai (2020) taking advantage of storage-related catchment properties and by Soulsby et al. (2021) for an experimental small-sized catchment. Across the study catchments, hydrological drought is characterized by concurrent negative P anomalies and positive PET anomalies, as shown for low flows in Switzerland by Floriancic et al. (2020), and generally attenuated compared to the corresponding meteorological drought by land surface processes (ET and ΔS), in agreement with Van Loon and Van Lanen (2012) (Fig. 4). Further, we showed that the storage-ET relationship during droughts is highly complex across a wide range of catchment types (Fig. 5a), depending on subsurface and vegetation properties (Mastrotheodoros et al., 2020). However, we argue that the storage behaviour, rather than that of ET only, drives the attenuation or intensification of Q anomalies compared to the P anomalies. In fact, catchments attenuating the hydrological drought generally experience concurrent negative ΔS anomalies (median ΔS anomaly = -112 mm for catchments attenuating the hydrological drought and = 81 mm for catchments intensifying it, Fig. 5b), while they can experience both positive and negative ET anomalies (median ΔS anomaly = -27 mm for catchments attenuating the hydrological drought and = 4 mm for catchments intensifying it, Fig. 5c). We argue that this is due to the different vegetation types in the catchments, their different ecophysiological properties and response to drought (Conte et al., 2019). Moreover, we found that catchments with contrasting subsurface storage-related properties show different behaviours in term of drought propagation (Fig. 6). This further suggests that catchments characterized by an inter-annual

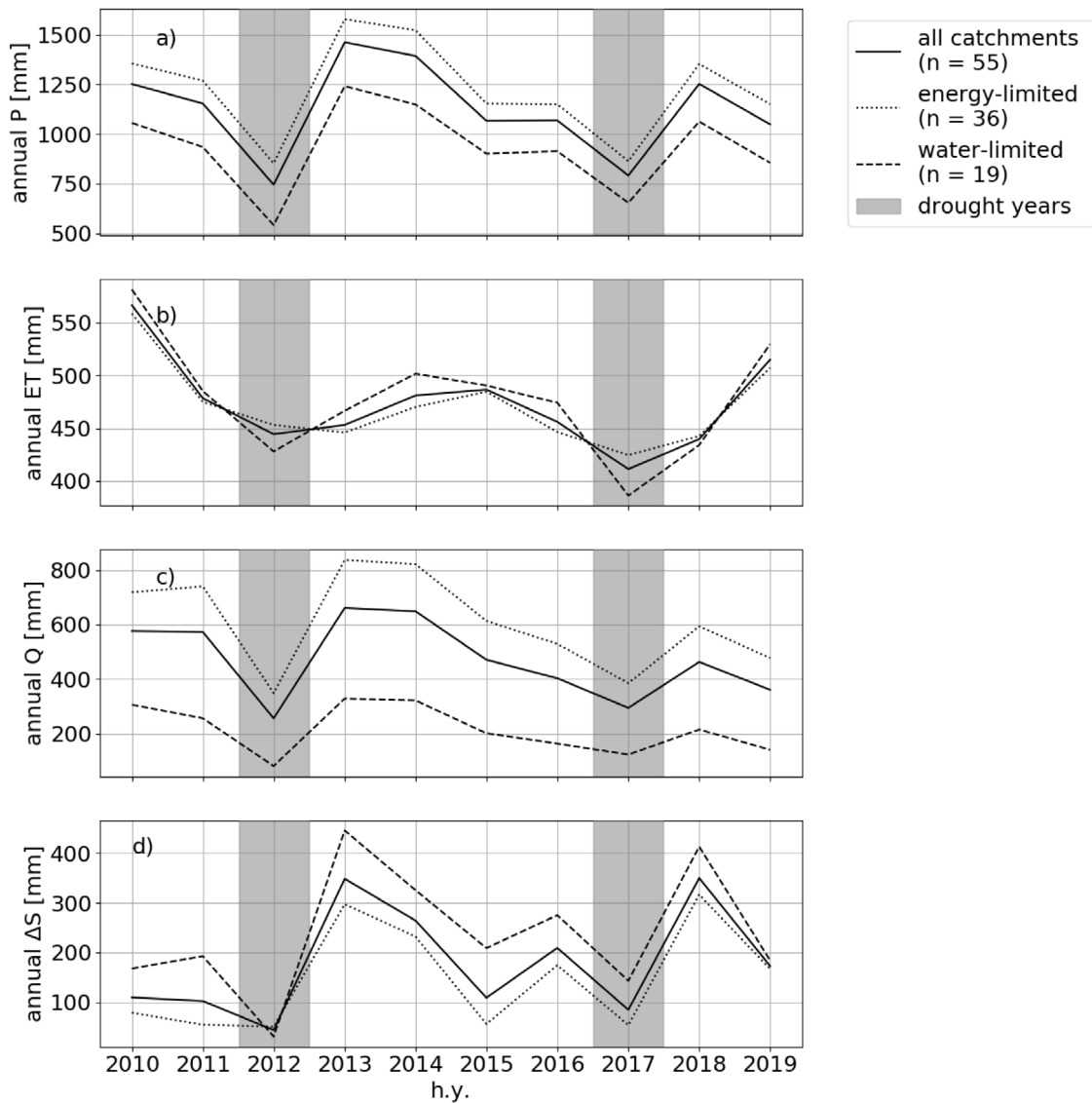


Fig. 8. Time series of annual (a) precipitation, P , (b) evapotranspiration, ET , (c) change in subsurface storage, ΔS , and (d) discharge, Q averaged across all catchments that experienced both the 2012 and 2017 droughts and non-drought conditions in the post-drought years (solid line, $n = 55$ catchments), water-limited (dashed line, $n = 19$ catchments) and energy-limited (dotted line, $n = 36$ catchments) catchments in the subset.

carryover of subsurface storage (as long-term mean or in the year preceding the drought) can attenuate the hydrological drought.

In a modelling experiment across the US (Apurv and Cai, 2020) found intensity and duration of hydrological droughts to be controlled by climatic properties, as well as the discharge sensitivity to storage, which they quantified from modelled baseflow and deep storage. Here we found that ASI can be thought as a predictor for the tendency of a catchment to attenuate or intensify the hydrological drought, quantified through the mean propagation (Fig. 7a), while we did not find a significant correlation between discharge sensitivity to storage and mean propagation (Fig. 7b). The difference we found with respect to Apurv and Cai (2020) may be due to the broader variety of discharge sensitivity to storage values across the catchments analysed in our study, as a result of the complex interplays among subsurface and climatic properties that determine the discharge sensitivity to storage of a catchment (Włostowski et al., 2021) (Fig. 7c). Furthermore, the quantification of discharge sensitivity to storage from recession data, as done here, relies on the assumption that catchments behave as simple first-order nonlinear dynamical systems (Kirchner, 2009), which

unavoidably may lead to a certain degree of uncertainty across a large-sample of catchments.

Third, we showed that Q recovery from P deficits takes place over similar time scales as ΔS , while ET has a buffered response (Fig. 8). Large-scale studies found a higher correlation between monthly Q and soil moisture – which is part of ΔS – rather than between other pairs of hydrometeorological variables at a continental and global scale (Ghajarnia et al., 2020b,a), and a delayed response of ET to soil moisture deficits across Europe (Orth and Destouni, 2018). Vegetation can have a delayed response to drought (Avanzi et al., 2020), because of biochemical and physiological processes (Conte et al., 2019). Most of the catchments we considered in the drought recovery analysis is dominated by two land cover types only (broad-leaved forest and crop), therefore we preferred to focus the analysis on water-limited versus energy-limited catchments, rather than on the dominant land cover type. However, different land cover types can be found within the study catchments, due to their generally relatively large scale. Thus, water-limited catchments can be thought as typical of Mediterranean

ecosystems, while the energy-limited ones of temperate and mountainous ecosystems (Table B.1), which can have different responses to dry periods because of different water use strategies (Conte et al., 2019). We indeed see a slight difference in ET response to drought between water- and energy-limited catchments: water-limited catchments have more pronounced ET variations during and after drought year compared to energy-limited catchments. We further hypothesize that the buffered response of ET to droughts can have more profound impacts on Q deficits in case of prolonged dry periods, and not just annual droughts as those analysed in this work. Some recent studies showed indeed that ET has a fundamental role in intensifying Q deficits compared to P deficits during multi-year droughts (Avanzi et al., 2020; Massari et al., 2022).

4.2. Data uncertainty

In this paper, we based our analyses on data products rather than on hydrological modelling, as hydrological models can miscapture the water balance during drought (Avanzi et al., 2020; Fowler et al., 2020). However, ET direct observation is not feasible at large scale and here we relied on an ET model forced by remote-sensed data, the LSASAF product (Section 2.2). To verify its suitability across the study region and during drought, we performed a multi-dataset comparison at flux tower- and catchment-scale (see Section 2.2 for a description of the ET products used for comparison). Monthly root mean square errors (RMSE) between LSASAF and FLUXNET2015 data over the comparison period (September 2009–December 2014) are below 30 mm (Fig. E.1a) and daily RMSE for the 2012 drought year below 1.5 mm (Fig. E.1c) across the selected flux towers, in agreement with previous validation exercises in Europe (Hu et al., 2015). Monthly catchment-scale ET data from LSASAF and alternative gridded products are highly correlated, for both the whole comparison period and drought months identified by the SPI12 ($r = 0.88$ for both the whole period and drought months between LSASAF and GLEAM, $r = 0.87$ for the whole period, and $r = 0.84$ for drought months between LSASAF and BIGBANG, Fig. E.1b and d). Hence, ET estimates from the selected remote-sensed product can be considered robust across the study catchments, over the whole spectrum of climatic conditions in the study period.

Since we assessed ΔS as annual residual from the water balance model (Eq. (1)), ΔS estimates could reflect uncertainties in P, Q, and ET data (Kampf et al., 2020). See Kampf et al. (2020) for a brief review about sources and magnitude of errors in the assessment of water balance components. Yet, we obtained comparable results in term of ΔS estimates using alternative P and ET data from the BIGBANG dataset ($r = 0.89$, Fig. F.1). Moreover, results from the correlation analysis between ΔS and catchment properties agree with previous literature (Section 4.1). Therefore, our ΔS estimates can be assumed as consistent and our general findings not dataset-dependent.

4.3. Future developments

In this work, we gained a consistent picture of the non-negligible annual change in subsurface storage, the attenuation of hydrological droughts due to subsurface storage depletion, and the similar time scale of discharge and subsurface storage recovery from drought conditions for 102 Italian catchments, regardless of the catchment and climatic properties. However, some study limitations pave the way for future research building on these results.

As a trade-off between data availability and variety in the characteristics of the considered catchments, we based our large-sample analysis on a 10-year time span. A longer record length would be desirable to include in the analysis additional drought events, potentially with different characteristics. Further, it would allow to use alternative methodologies to study drought propagation, as the threshold level method (Van Loon and Van Lanen, 2012) or a standardized approach (Barker et al., 2016) through the quantification of indexes for

each hydrological compartment, such as the Standardized Precipitation Evapotranspiration Index (SPEI) (Vicente-Serrano et al., 2010), the Palmer Drought Severity Index (PDSI) (Palmer, 1965), the Standardized Streamflow Index (SSI) (Vicente-Serrano et al., 2012), the Standardized Soil Moisture Index (SSMI) (Carrão et al., 2016), and the Standardized Groundwater Index (SGI) (Bloomfield and Marchant, 2013).

Moreover, we estimated ΔS as annual water balance closure, whereas long-term soil moisture and groundwater data would allow to directly track storage changes either in the soil and in the aquifers, and thus to investigate their different role in drought evolution (Teuling et al., 2013), as well as to quantify the memory effect in groundwater levels (Mangin, 1984) and its relationship with drought propagation (Bloomfield and Marchant, 2013) in the study region.

Finally, we did not consider anthropogenic activities (e.g., surface- and ground-water abstractions, reservoir infrastructures, land use changes, and water transfers) that may occur in the study catchments and may affect their response to drought (Van Loon et al., 2016; Wu et al., 2019; Margariti et al., 2019; Van Loon et al., 2022). (Van Loon et al., 2022) showed that human activities aggravated hydrological drought characteristics compared to natural conditions in 28 human-altered catchments across the world, regardless of the typology and purpose of these activities. Given the consistent variation observed in hydrological drought characteristics due to various human activities (Van Loon et al., 2022), we argue that the neglect of possible anthropogenic disturbances in the study catchments does not affect our findings. However, detailed information about human activities in the study catchments would be beneficial to test for instance if the intensification of the hydrological drought we observed in some catchments can be related to anthropogenic influences, which was beyond the scope of the present work.

5. Conclusions

Hydrological drought is shaped by several factors, including the interaction between meteorological drought, i.e., precipitation deficits and possibly increases in temperature and potential evapotranspiration, and land surface processes, e.g., actual evapotranspiration and storage in the soil and weathered bedrock. Nevertheless, systematic data-based analyses of the response of each water balance component (precipitation, discharge, evapotranspiration, and subsurface storage change) to drought across different climates and catchment types are rare. Here, we revealed that such analyses are feasible blending ground-based and remote-sensed data. Specifically, we showed that long-term mean annual subsurface storage change is on average 11% of mean annual precipitation across 102 Italian catchments along a steep climatic and topographic gradient (Fig. 3). Thus, neglecting subsurface storage change could bias water balance assessments at annual scale. Furthermore, we pointed out that hydrological drought is generally attenuated compared to meteorological drought (Fig. 4) and this is mainly due to a net depletion of subsurface storage (Figs. 5 and 6) across a large set of catchments with different morphological properties and in different hydroclimatic regimes. Finally, we showed that across the study catchments subsurface storage and discharge have a similar timing in recovery from precipitation deficit, while evapotranspiration has a buffered response to dry periods (Fig. 8). Our results reveal the usefulness of explicitly considering subsurface storage changes to disentangle hydrological processes during drought and provide opportunities to properly inform water managers about water availability in the different hydrological compartments during such periods.

CRedit authorship contribution statement

Giulia Bruno: Methodology, Software, Formal Analysis, Investigation, Writing – original draft, Visualization. **Francesco Avanzi:** Conceptualization, Methodology, Writing- Original draft preparation. **Simone Gabellani:** Conceptualization, Writing – reviewing and editing.

Luca Ferraris: Supervision, Writing – reviewing and editing. **Edoardo Cremonese:** Investigation, Writing – reviewing and editing. **Marta Galvagno:** Investigation, Writing – reviewing and editing. **Christian Massari:** Conceptualization, Writing – reviewing and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

The Asynchronicity Index (ASI) between precipitation (P) and potential evapotranspiration (PET) (Feng et al., 2019) is computed as

$$ASI = \sqrt{JS_{obs} - JS_{sim}}, \quad (A.1)$$

where JS_{obs} is the observed Jensen–Shannon distance and JS_{min} is the minimum Jensen–Shannon distance that can be obtained for the seasonality of P and PET. The Jensen–Shannon distances (JS) are derived as

$$JS = \sqrt{\frac{1}{2}D_P + \frac{1}{2}D_{PET}} \quad (A.2)$$

where D_P is the relative entropy of P and D_{PET} is the relative entropy of PET with reference to their average, obtained as

$$D_P = \sum_{m=1}^{12} P_m \log_2 \frac{P_m}{n_m} \quad (A.3)$$

and

$$D_{PET} = \sum_{m=1}^{12} PET_m \log_2 \frac{PET_m}{n_m} \quad (A.4)$$

with

$$P_m = \frac{r_m}{\sum_{m=1}^{12} r_m} \quad (A.5)$$

$$PET_m = \frac{k_m}{\sum_{m=1}^{12} k_m} \quad (A.6)$$

$$n_m = \frac{1}{2} (P_m + PET_m) \quad (A.7)$$

where r_m and k_m are the long-term mean monthly values for P and PET and the calendar month $m \in [1, 12]$.

Appendix B

Table B.1

Catchment properties: basin and section name, catchment mean latitude, catchment area in km², catchment mean elevation in m a.s.l., aridity index (AI), dominant land cover type in the catchment, dominant topsoil texture in the catchment, and dominant geological class in the catchment. Sources and details about each property are provided in Section 2.2.2. Catchments are north to south ordered, according to the catchment mean latitude.

BASIN	SECTION	MEAN LAT	AREA	ELEV	AI	LAND COVER	SOIL TEXTURE	GEOLOGICAL CLASS
Adige	S Michele all Adige	46.71	7299	1970	0.52	Coniferous forest	Medium	Igneous and metamorphic
Adige	Trento	46.64	9709	1935	0.55	Coniferous forest	Medium	Igneous and metamorphic
Adige	Verona	46.58	11029	1828	0.75	Coniferous forest	Medium	Igneous and metamorphic
Adige	Boara Pisani	46.52	12203	1708	0.63	Coniferous forest	Medium	Unconsolidated
Rabbies	S Bernardo Rabbi	46.43	105	2487	0.62	Bare soil	Medium	Igneous and metamorphic
Avisio	Cavalese Masi	46.38	569	2117	0.66	Coniferous forest	Medium	Unconsolidated
Noce	Mezzolombardo	46.37	1349	1870	0.58	Coniferous forest	Medium	Unconsolidated
Noce	Male	46.33	467	2288	0.55	Coniferous forest	Medium	Unconsolidated
Avisio	Lavis	46.3	933	1886	1.02	Coniferous forest	Medium	Unconsolidated
Simone	Fiera di Primiero	46.25	124	1899	0.47	Coniferous forest	Coarse	Unconsolidated
Toce	Domodossola	46.24	828	1953	0.65	Bare soil	Coarse	Igneous and metamorphic
Sarca	Ragoli	46.16	539	2077	0.86	Coniferous forest	Medium	Igneous and metamorphic
Adda	Lodi	46.16	5584	1542	0.58	Broad-leaved forest	Medium	Unconsolidated
Toce	Candoglia	46.12	1441	1845	0.32	Bare soil	Coarse	Igneous and metamorphic
Fersina	Trento Fersina	46.11	174	1315	0.8	Coniferous forest	Coarse	Igneous and metamorphic
Brenta	Barzizza	46.08	1564	1473	0.35	Coniferous forest	Medium	Carbonate
Brenta	Grigno Ponte Filippini	46.06	616	1457	0.53	Coniferous forest	Medium	Igneous and metamorphic
Chiese	Cimego	46.01	238	2046	0.81	Coniferous forest	Medium	Igneous and metamorphic
Oglio	Capriolo	45.99	1883	1591	0.79	Coniferous forest	Medium	Unconsolidated
Chiese	Ponte dei Tedeschi	45.98	369	1841	0.43	Coniferous forest	Medium	Unconsolidated
Ticino	Vigevano	45.96	7071	1179	0.24	Broad-leaved forest	Medium	Igneous and metamorphic
Tesina	Bolzano Vicentino	45.85	809	1173	0.94	Coniferous forest	Medium	Carbonate
Ticino	Ponte della Libertà	45.85	8018	1018	0.28	Broad-leaved forest	Medium	Unconsolidated
Chiese	Gavardo	45.84	934	1430	0.45	Broad-leaved forest	Medium	Unconsolidated
Olona	Castellanza Olona	45.79	163	607	0.61	Broad-leaved forest	Medium	Unconsolidated
Lambro	Peregallo	45.79	273	654	0.63	Broad-leaved forest	Medium	Unconsolidated
Dorabatea	Tavagnasco	45.75	3322	2300	0.44	Bare soil	Medium	Igneous and metamorphic
Bacchiglione	Montegalda	45.74	1470	844	0.65	Broad-leaved forest	Medium	Carbonate
Oglio	Marcara	45.72	5840	979	0.58	Crops	Medium	Unconsolidated
Dorabatea	Verolengo	45.68	3940	2063	0.67	Bare soil	Medium	Igneous and metamorphic
Agno	Ponte Brogliano	45.68	143	849	0.52	Broad-leaved forest	Medium	Silicatic
Sesia	Quinto Vercellese Cervo	45.56	1055	702	0.54	Broad-leaved forest	Medium	Unconsolidated
Agno	Borgofrassine	45.56	526	561	0.77	Broad-leaved forest	Medium	Unconsolidated
Orco	S.Benigno	45.45	733	1982	0.46	Bare soil	Medium	Igneous and metamorphic
Po	Cremona	45.43	50403	1064	0.56	Broad-leaved forest	Medium	Unconsolidated
SturadiLanzo	Lanzo	45.29	578	1992	0.42	Bare soil	Coarse	Igneous and metamorphic
Po	Ficarolo	45.24	68619	977	0.59	Crops	Medium	Unconsolidated
Po	Casale Monferrato	45.19	13715	1478	0.73	Crops	Medium	Igneous and metamorphic
Po	Spessa	45.13	37500	1085	0.55	Broad-leaved forest	Medium	Unconsolidated
DoraRiparia	Torino	45.1	1416	1768	0.63	Broad-leaved forest	Medium	Igneous and metamorphic
DoraRiparia	Susa Via Mazzini	45.06	652	2273	0.53	Coniferous forest	Medium	Igneous and metamorphic
Po	Torino Murazzi	44.77	5177	1145	0.66	Crops	Medium	Unconsolidated
Po	Carignano	44.69	3785	1364	0.61	Crops	Medium	Igneous and metamorphic
Trebbia	Rivergaro	44.68	890	1060	0.56	Broad-leaved forest	Medium	Silicatic
Crosto	Cadelbosco	44.65	257	426	0.99	Crops	Fine	Unconsolidated
Taro	S.Secondo	44.63	1488	836	0.56	Broad-leaved forest	Medium	Unconsolidated
Maira	Busca	44.5	580	1895	0.79	Bare soil	Medium	Igneous and metamorphic
Secchia	Pioppa	44.5	1631	774	0.72	Broad-leaved forest	Medium	Unconsolidated
Erro	Cartosio	44.5	199	531	0.65	Broad-leaved forest	Medium	Igneous and metamorphic
Secchia	Ponte Alto	44.44	1420	852	0.74	Broad-leaved forest	Medium	Unconsolidated
Bormida	Piana Crixia	44.38	279	535	1	Broad-leaved forest	Medium	Igneous and metamorphic
Rossenna	Rossenna	44.37	183	940	0.8	Broad-leaved forest	Medium	Unconsolidated
Tanaro	Alba Q.A.	44.35	3437	1281	0.6	Broad-leaved forest	Medium	Igneous and metamorphic
Panaro	Bomporto	44.34	1082	823	0.69	Broad-leaved forest	Medium	Unconsolidated
Tanaro	Farigliano	44.32	1517	1158	0.52	Broad-leaved forest	Medium	Igneous and metamorphic
Scoltenna	Pievepelago	44.2	126	1530	0.38	Broad-leaved forest	Medium	Silicatic
Reno	Vergato	44.17	535	785	0.52	Broad-leaved forest	Medium	Silicatic
Neva	Cisano sul Neva	44.13	128	741	0.69	Broad-leaved forest	Medium	Carbonate
Serchio	Calavorno	44.13	681	919	0.71	Broad-leaved forest	Medium	Silicatic
Lamone	Strada Casale	44.09	200	656	0.85	Broad-leaved forest	Medium	Silicatic
Serchio	Monte S.Quirico	44.09	1254	836	0.5	Broad-leaved forest	Medium	Silicatic
Serchio	Ripafra	44.07	1401	769	0.48	Broad-leaved forest	Medium	Silicatic
Uso	Santarcangelo	44	112	307	0.96	Crops	Medium	Silicatic
Arno	S. Piero a Ponti	43.97	287	519	0.83	Broad-leaved forest	Medium	Silicatic
Arno	Fornacina 2 ul	43.95	828	563	0.82	Broad-leaved forest	Medium	Silicatic
Arno	Poggio a Caiano	43.94	440	352	0.86	Broad-leaved forest	Medium	Unconsolidated

(continued on next page)

Table B.1 (continued).

BASIN	SECTION	MEAN LAT	AREA	ELEV	AI	LAND COVER	SOIL TEXTURE	GEOLOGICAL CLASS
Savio	S.Carlo	43.92	578	586	0.72	Broad-leaved forest	Medium	Silicatic
Casentino	Subbiano	43.73	750	817	0.72	Broad-leaved forest	Medium	Silicatic
Arno	Ponte di Scandicci	43.67	275	362	1.5	Crops	Medium	Silicatic
Arno	S.Giovanni alla Vena valle	43.58	8601	416	1.06	Broad-leaved forest	Medium	Silicatic
Tevere	SLucia	43.55	937	640	0.86	Broad-leaved forest	Medium	Silicatic
Aspio	Aspio 2	43.54	134	143	1.39	Crops	Medium	Unconsolidated
Carpina	Montone	43.42	133	601	1.19	Broad-leaved forest	Medium	Silicatic
Arno	Incisa Valle	43.41	3414	513	1.05	Broad-leaved forest	Medium	Silicatic
Tevere	Mocaiana	43.39	108	659	1.04	Crops	Medium	Silicatic
Nestore	Trestina	43.35	204	572	1.05	Broad-leaved forest	Medium	Silicatic
Chiascio	Branca	43.32	242	712	0.81	Crops	Medium	Silicatic
Niccone	Migianella	43.27	134	514	1.05	Broad-leaved forest	Medium	Silicatic
Ombrone	Sasso d'Ombrone	43.12	2706	425	1.19	Broad-leaved forest	Medium	Silicatic
Topino	Valtopina	43.11	196	745	0.81	Broad-leaved forest	Medium	Silicatic
Tevere	Alviano	43.03	7438	524	1.21	Broad-leaved forest	Medium	Silicatic
Puglia	Collepepe	42.91	162	443	1.25	Crops	Medium	Silicatic
Nera	Ponte Buggianino	42.88	396	1227	1.02	Broad-leaved forest	Medium	Carbonate
Carcaione	Orvieto Scalo	42.83	1306	512	1.07	Broad-leaved forest	Medium	Unconsolidated
Tevere	Cantalupo	42.81	554	550	1.24	Crops	Medium	Carbonate
Tevere	Vallo di Nera	42.79	1238	1126	1.1	Broad-leaved forest	Medium	Carbonate
Tevere	Ponte Felice	42.75	12896	660	0.95	Broad-leaved forest	Medium	Silicatic
Tevere	Torreorsina	42.75	1467	1080	1.07	Broad-leaved forest	Medium	Carbonate
Tevere	Ripetta	42.64	16595	610	1.01	Broad-leaved forest	Medium	Silicatic
Nera	Serravalle	42.64	429	1160	0.98	Broad-leaved forest	Medium	Carbonate
Fiora	Montalto di Castro	42.61	824	421	1.21	Crops	Medium	Igneous and metamorphic
Marta	Tarquinia	42.41	1070	340	1.59	Crops	Coarse	Igneous and metamorphic
Velino	Terria	42.26	2194	1036	0.65	Broad-leaved forest	Medium	Carbonate
Turano	Rocca Sinibalda	42.08	503	1039	0.92	Broad-leaved forest	Medium	Silicatic
Aniene	Lunghezza	41.93	1201	673	0.92	Broad-leaved forest	Medium	Carbonate
Aniene	Ponte Mammolo	41.92	1494	586	0.7	Broad-leaved forest	Medium	Igneous and metamorphic
Liri	Pontecorvo	41.76	3759	737	0.86	Broad-leaved forest	Medium	Carbonate
Melfa	Atina	41.66	169	1071	0.84	Broad-leaved forest	Medium	Carbonate
Rapido	Cassino	41.57	189	627	0.62	Broad-leaved forest	Medium	Carbonate
Ofanto	San Samuele	40.99	2697	499	1.36	Crops	Medium	Silicatic
Bradano	Ponte Bradano	40.75	2999	429	1.52	Crops	Medium	Silicatic
Agri	Ponte La Marmora	40.37	276	975	0.86	Broad-leaved forest	Medium	Silicatic

Appendix C

Table C.1
Shapiro–Wilk rejection frequency for each candidate distribution for monthly SPI12 calculation across the study catchments.

Gamma	Gumbel	Logistic	Loglogistic	Lognormal	Normal	Weibull
0.62	10.07	20.19	5.31	0.96	12.01	0.87

Appendix D

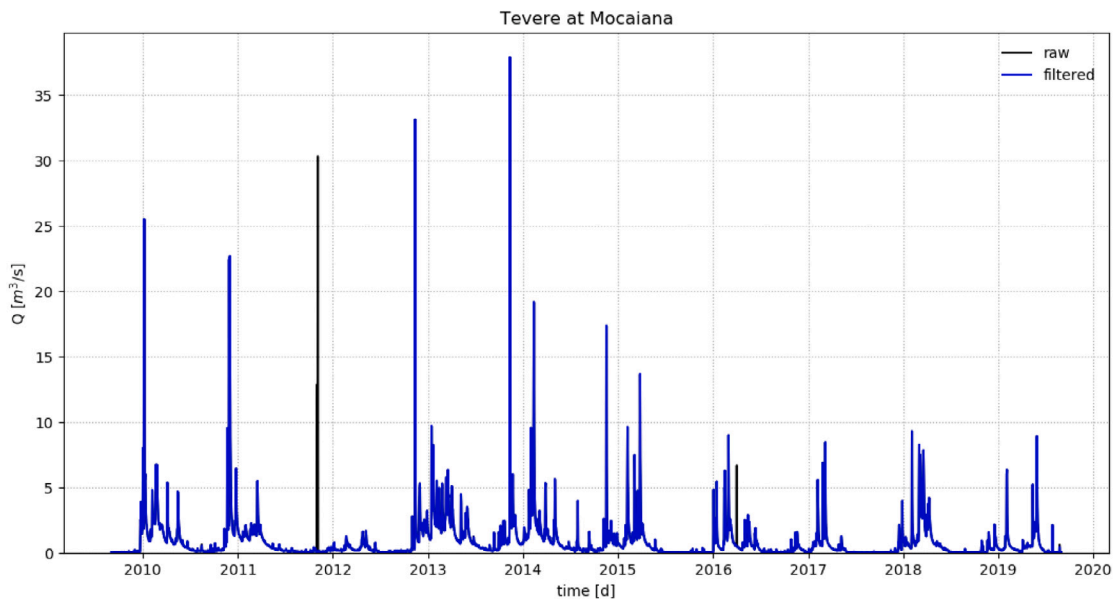


Fig. D.1. Example of raw and filtered mean daily discharge (Q) data, according to the quality-check procedure described in Section 2.2.1.

Appendix E

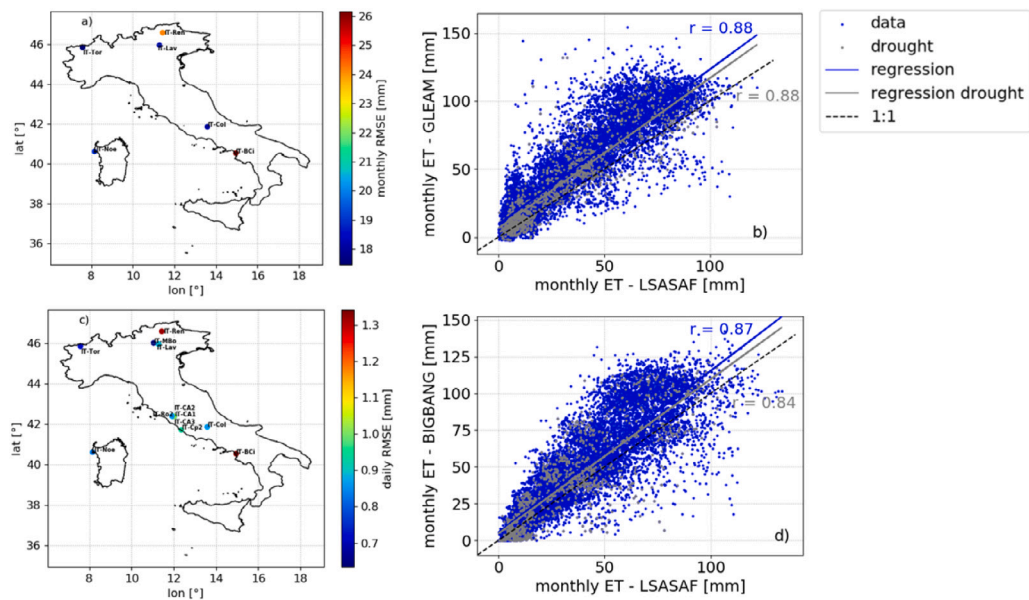


Fig. E.1. Multi-dataset evapotranspiration (ET) comparison across the study region: (a) monthly root mean square error (RMSE) over the whole comparison period and (c) daily RMSE for the 2012 drought year between ET data from FLUXNET2015 dataset and LSASAF product at flux tower-scale (site IDs, according to the FLUXNET2015 dataset, are reported); (b) scatterplot between monthly catchment-average ET from LSASAF and GLEAM datasets over the whole comparison period (grey) and drought months (red); (d) same as (b) with catchment-average ET from the BIGBANG dataset on the y axis. Description of datasets is provided in Section 2.2.1. In (b) and (d), Pearson correlation coefficients (r) are reported.

Appendix F

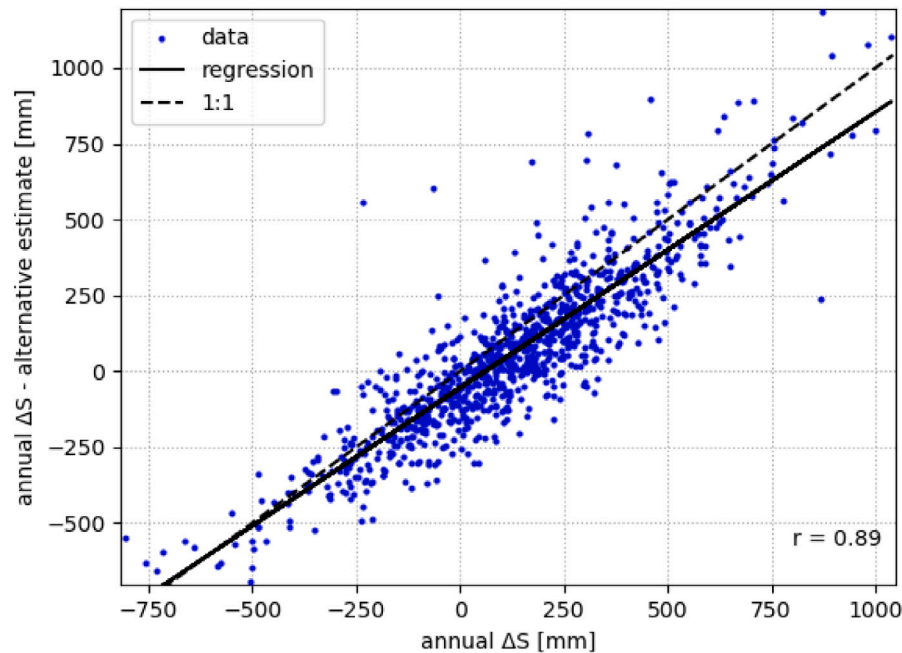


Fig. F.1. Scatterplot between annual change in subsurface storage (ΔS) estimates used in this study and alternative estimates. Alternative ΔS estimates are annual residual from precipitation and evapotranspiration data from the BIGBANG dataset, and discharge data collected in this study (Section 2.2.1), according to the water balance equation (Eq. (1)). Pearson correlation coefficient (r) is reported.

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